A Knowledge-Based Approach to UAV Path Planning for Bridge Maintenance in a Digital Twin Environment

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Abstract: Unmanned aerial vehicles (UAVs) are increasingly used for bridge inspection to reduce hazardous access and improve scalability. Routes are often manual or coverage-driven, and the collected imagery is not always comparable or aligned with maintenance decision needs. This study proposes a knowledge-based UAV path planning approach that converts maintenance intent into executable and repeatable inspection routes. The method integrates element-level maintenance knowledge from field inspection and maintenance records encoded as priority weights and viewing requirements with BIM derived bridge geometry to generate candidate viewpoints and element specific inspection patterns for substructure and superstructure. Viewpoints are evaluated using a maintenance-weighted utility function. A case study on a conventional concrete box-girder toll-road bridge compares the approach with three baselines manual rule-of-thumb, a coverage only planner, and a robustness oriented feasibility proxy. Results show routes that remain traceable to maintenance priorities and support cycle-to-cycle repeatability for continuous inspection planning.

Keywords: Bridge Inspection; Bridge Maintenance; UAV Path Planning; Knowledge-Based Planning; BIM; Digital Twin; Viewpoint Selection; Repeatable Inspection Routes

I. Introduction

Bridges are critical components of transportation networks, and their long-term performance strongly depends on effective inspection and maintenance strategies. Aging infrastructure, increasing traffic demand, and environmental exposure have significantly intensified the need for reliable and timely bridge condition assessment (Agredo Chávez et al., 2024; Mishra et al., 2022). Conventional bridge inspections are predominantly based on manual visual surveys, which are time-consuming, costly, and often limited by accessibility and safety constraints, particularly for hard-to-reach components such as bridge decks, girders, bearings, and piers (Gucunski et al., 2015; Ye, 2020, 2022).

In recent years, unmanned aerial vehicles (UAVs) have emerged as a promising alternative for bridge inspection, offering improved accessibility, reduced inspection time, and enhanced safety for inspectors (Aliyari, 2022; Ayele, 2020). Equipped with high-resolution cameras and stabilized platforms, UAVs enable close-range visual inspection without the need for scaffolding, traffic closures, or specialized access equipment (Ameli, 2022; Biscarini, 2020). Consequently, UAV-based inspection has been increasingly adopted in both research and practice for bridge condition assessment and monitoring.

Alongside the growing use of UAVs, significant research efforts have focused on automated damage detection using computer vision and artificial intelligence techniques. Deep learning-based methods have demonstrated strong capabilities in detecting cracks, spalling, corrosion, and other surface defects from UAV-acquired images (Hu et al., 2024; Kao, 2023; Luciw et al., 2024). While these approaches

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have substantially improved post-processing and interpretation of inspection data, they largely assume that inspection data are already available and of sufficient quality. The flight planning process that determines where and how inspection data are collected often remains manual or only loosely defined.

Existing UAV flight planning strategies for infrastructure inspection are commonly based on generic coverage path planning or viewpoint optimization principles originally developed for mapping, photogrammetry, or robotic exploration tasks (Aela, 2024; Fang, 2024; Hildmann & Kovacs, 2019). These methods typically aim to maximize surface coverage or minimize flight time and energy consumption. Although effective for geometric data acquisition, such approaches do not explicitly account for bridge maintenance objectives, such as the prioritization of critical structural components or the need for consistent inspection viewpoints across multiple inspection cycles (Falorca, 2021; Ri, 2024).

More recent studies have attempted to tailor UAV path planning to bridge inspection scenarios by considering environmental constraints, complex geometries, or specific structural configurations. For example, research has addressed UAV navigation under bridge decks with limited GNSS availability (Shults et al., 2023), wind-aware path planning for long-span bridges (Xu, 2021), and surface-specific inspection of bridge substructures (Omar & Nehdi, n.d.). Despite these advances, most existing approaches remain predominantly geometry-driven, focusing on obstacle avoidance and surface coverage rather than inspection relevance from a maintenance perspective.

Parallel to developments in UAV inspection, the field of bridge maintenance has seen increasing interest in knowledge-based systems and decision support tools. Knowledge graphs, ontologies, and data-driven frameworks have been proposed to organize maintenance knowledge, defect characteristics, and decision rules in a structured manner (Gao et al., 2024; Hoang, 2021). More recently, advanced reasoning frameworks integrating large language models with bridge maintenance knowledge graphs have demonstrated the potential to support expert-level maintenance decision-making (Wang et al., 2026). These studies highlight the importance of domain knowledge in guiding maintenance actions but generally focus on diagnosis and recommendation rather than the operational execution of inspections.

As a result, a critical gap remains between maintenance knowledge and UAV-based inspection planning. Current UAV inspection workflows rarely incorporate maintenance knowledge, such as component criticality, historical damage patterns, or inspection requirements, into the path planning process. Consequently, inspection routes are often inconsistent across inspection cycles, limiting their usefulness for long-term monitoring and maintenance planning (Yeum & Dyke, 2015).

To address this gap, this study proposes a knowledge-based approach to UAV path planning for bridge inspection and maintenance. The proposed approach integrates maintenance knowledge with bridge geometry to define inspection objectives and generate UAV flight paths that are inspection-oriented, maintenance-relevant, and repeatable. By shifting the focus from generic coverage to knowledge-informed inspection planning, the proposed framework aims to support continuous bridge inspection and provide a practical foundation for long-term maintenance-oriented UAV deployment.

II. Background and Related Work

2.1 UAV-Based Bridge Inspection in a Maintenance Context

Over the last five years, UAV-based bridge inspection has increasingly been discussed not only as a safer alternative to conventional access methods, but also as a pathway to make inspections faster and more scalable. Recent application reviews report that drones can capture close-range imagery for components that are otherwise difficult to access—such as the underside of decks, substructures, and elevated superstructure elements—while reducing the need for traffic closures or risky manual access

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(Panigati et al., 2025). This gradual shift from ad hoc deployment toward more routine use is reflected in the recent trajectory of the field, where attention has moved from demonstrating feasibility to improving consistency and operational readiness (see Figure 1).

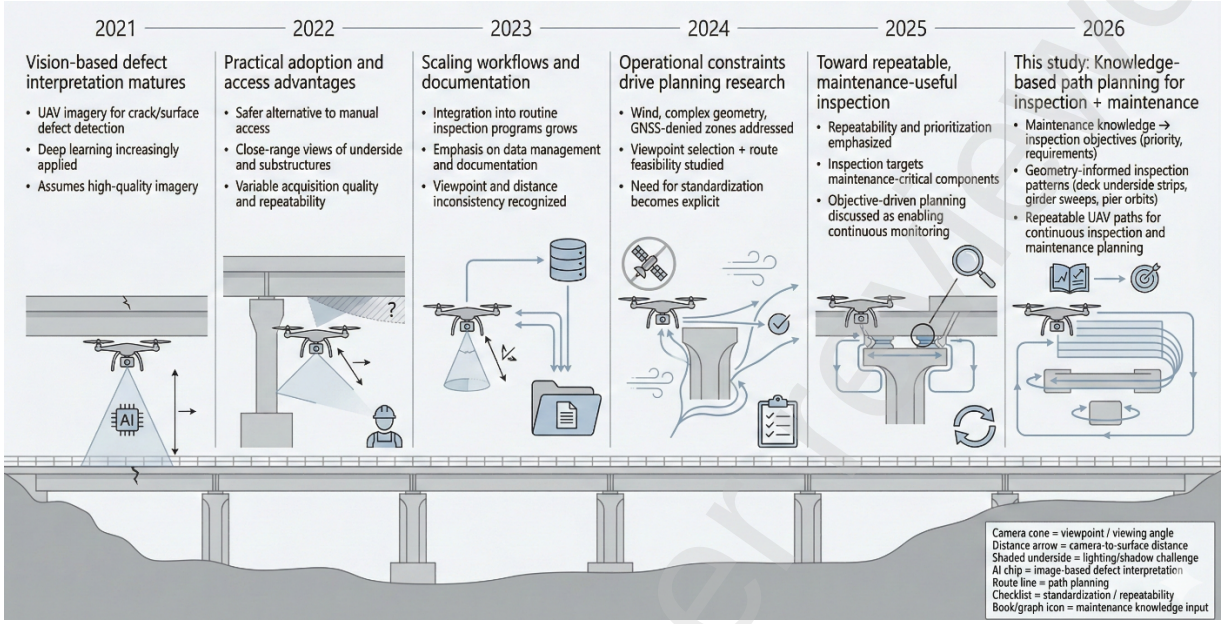


Fig. 1. Evolution of UAV-based bridge inspection over the last five years

At the same time, the literature makes it clear that inspection value is determined by more than the mere availability of drone imagery. Several studies emphasize that UAV inspection outcomes can vary widely because of differences in flight execution, camera-to-surface distance, viewing angles, and lighting conditions, particularly around shaded regions or complex geometries (Hou et al., 2022; Panigati et al., 2025). These findings highlight a practical challenge that is often understated in technology-driven demonstrations: unless data acquisition is performed in a consistent manner, it becomes difficult to compare condition information across inspection cycles and to use UAV outputs reliably for maintenance planning.

A parallel stream of research has focused on image-based defect interpretation, where deep learning and related computer vision techniques are used to identify cracks and surface damage from captured imagery. This line of work has produced promising results and continues to mature rapidly (Aliyari et al., 2021). However, these advances typically assume that the input imagery already satisfies inspection needs in terms of resolution, visibility, and coverage. In practice, the quality and relevance of the data depend heavily on how the drone is guided during inspection, including whether the UAV repeatedly captures comparable viewpoints and meets inspection-specific viewing requirements. As a result, recent reviews increasingly point to flight planning and operational standardization as limiting factors for turning UAV inspection from a one-off assessment into a repeatable maintenance-support workflow (Hou et al., 2022; Panigati et al., 2025).

In maintenance settings, repeatability and prioritization become especially important. Unlike exploratory surveys, bridge inspections are expected to support consistent condition tracking over time and to focus effort where it matters most for maintenance decisions. The emerging consensus is that UAV inspection will be most valuable when it can consistently capture comparable viewpoints and sufficiently detailed observations for maintenance-critical components, rather than simply maximizing generic surface coverage (Hou et al., 2022; Toriumi et al., 2022). This maintenance perspective

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motivates the need for planning approaches that move beyond manual route definition and toward more systematic, objective-driven flight planning—an evolution that the 2021–2025 literature increasingly anticipates and that this study addresses through a knowledge-based planning perspective (Figure 1).

2.2 UAV Path Planning for Bridge Inspection

Research on UAV path planning for bridge inspection has grown alongside the wider adoption of drones in infrastructure monitoring. Much of this work starts from a practical observation: even when a drone platform and camera are suitable for inspection, the usefulness of the collected data depends strongly on where the UAV flies, how it approaches each component, and how consistently viewpoints can be reproduced. As a result, many recent studies formulate path planning as a combination of viewpoint selection and route optimization, typically aiming to reduce redundant observations while maintaining adequate coverage and feasible maneuvering around complex bridge geometries (Li et al., 2025).

Several contributions have emphasized coverage- and efficiency-oriented planning, where the primary objective is to maximize inspected surface area or minimize flight time and path length. This direction is often motivated by limited battery endurance and the need to complete inspections within operational constraints. In bridge contexts, the focus has been placed on distributing viewpoints to reduce overlap and avoid unnecessarily sharp turns, which can improve both flight stability and practical field execution (Li et al., 2025). Such approaches provide important foundations for operational feasibility, especially when inspection tasks are framed as surface coverage problems.

Another line of work has sought to address bridge-specific operational challenges that cannot be captured by generic coverage formulations. Environmental disturbance, particularly wind effects, has been incorporated into planning and evaluation to improve route realism for long-span bridges and exposed conditions (Xu & Dai, 2024). In addition, the GNSS shadowing that frequently occurs beneath bridge decks and near substructures has motivated planning strategies that account for navigation limitations and operational modes, recognizing that feasible inspection routes may require different behaviors under degraded positioning conditions (Wang, 2024). These studies improve robustness by acknowledging that bridges are not open, ideal environments for UAV navigation.

2.3 Knowledge-Based Bridge Maintenance, Gaps, and Research Positioning

Bridge maintenance research increasingly uses knowledge graphs and ontologies to structure components, defects, inspection records, and maintenance actions for more consistent retrieval and decision support (Zhang et al., 2023). Recent studies extend this by enriching maintenance knowledge from textual records and formalizing actionable preservation rules, while LLM-assisted reasoning has been introduced to emulate expert workflows over maintenance knowledge graphs (Gao et al., 2024; He et al., 2024; Wang et al., 2026). However, these approaches still largely produce diagnosis or recommendations rather than executable inspection actions. This leaves a clear gap in translating maintenance knowledge into inspection objectives and repeatable UAV routes for continuous inspection and maintenance planning, as summarized in Table 1.

Table 1. Systematic Mapping of Recent Studies and the Research Gap

Cluster	Representative studies	Primary focus	Mechanism	Key limitation for maintenance-oriented UAV planning
UAV bridge inspection practice & reviews	Panigati et al. (2025); Toriumi et al. (2022)	Workflows, limitations, adoption	Review synthesis	Repeatable routing and maintenance linkage not fully formalized

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UAV + vision-based defect analysis	Aliyari et al. (2021)	Defect detection from UAV imagery	Deep learning	Assumes "good" inspection data; planning stage often external/manual
Bridge coverage/viewpoint optimization	Li et al. (2025)	Reduce redundancy & improve efficiency	Viewpoint optimization + routing	Maintenance intent (criticality/requirements) not explicit
Environmental-aware path planning	Xu & Dai (2024)	Wind effects on feasible routes	Wind-aware cost modeling	Mostly feasibility/efficiency-driven; limited maintenance prioritization
GNSS-denied substructure inspection	Kim & Youn (2025)	Under-bridge GNSS shadowing	Operational-mode planning	Focus on navigation feasibility; inspection objectives still generic
Bridge maintenance knowledge base (KG)	Zhang et al. (2023)	Structuring maintenance knowledge	Knowledge graph	Focus on representation/retrieval; not tied to UAV execution
Knowledge-driven maintenance decisions	Jiang et al. (2023)	Holistic maintenance support	Semantic reasoning + ML links	Operational inspection execution not addressed
LLM-assisted maintenance reasoning	Wang et al. (2026)	Expert-like reasoning over maintenance KG	KG + LLM CoT reasoning	Emphasis on recommendation rather than executable UAV inspection plans
Maintenance-informed UAV path planning	**Laksono et al. (2026)**	**Knowledge-based UAV path planning for bridge inspection and maintenance**	**Maintenance knowledge → inspection objectives; geometry-informed inspection patterns; repeatable route generation**	**Addresses the translation gap by producing executable, repeatable UAV inspection paths aligned with maintenance priorities**

UAV inspection and planning studies have improved feasibility and robustness in bridge environments, addressing factors such as wind, GNSS shadowing, and viewpoint redundancy, and some work has linked UAV trajectories to digital-twin or bridge management workflows (Bono et al., 2022). However, these approaches rarely formulate planning as the translation of maintenance knowledge into explicit inspection objectives, so routes are still largely derived from geometric and operational considerations while maintenance knowledge remains separate. Taken together, the literature points to a gap in connecting knowledge-based maintenance systems with UAV path planning through an explicit pipeline that maps component criticality, inspection requirements, and condition history into executable and repeatable routes. This study addresses that gap through a knowledge-based planning approach that uses maintenance intent to define inspection objectives and combines it with bridge geometry and UAV constraints to generate feasible, element-appropriate routes that remain consistent across inspection cycles.

III. PROPOSED FRAMEWORK

3.1 Framework Overview

This study proposes a knowledge-based framework for UAV path planning that is intended to support bridge inspection as a repeatable input to maintenance planning. The framework is designed around a simple premise: inspection routes should not be derived solely from geometric coverage or operational convenience, but should reflect maintenance intent by prioritizing maintenance-critical components and meeting inspection-specific viewing requirements. Figure 2 illustrates the overall workflow and the information flow between its main elements.

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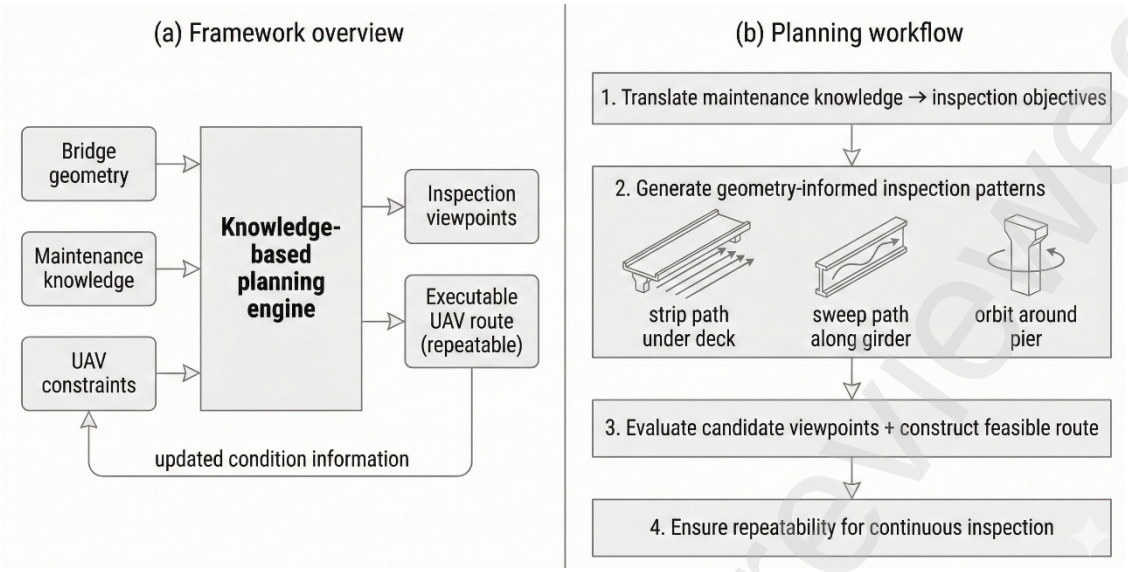


Fig. 2. Overview Proposed Framework

The framework takes three primary inputs. First, it requires a representation of bridge geometry that enables element-level inspection planning, such as decks, girders, piers, and other substructure regions. This geometry can be obtained from different sources depending on data availability, including a simplified structural model, a surface model derived from point clouds, or geometry extracted from a BIM environment; the framework does not depend on a single data format as long as the geometry can be represented in an inspection-relevant way. Second, the framework incorporates maintenance knowledge as prior information that captures inspection priorities and requirements. In practice, this includes component criticality, historical condition or damage patterns, and element-specific requirements related to camera-to-surface distance, viewing angle, and expected observation quality. Third, UAV operational constraints are considered to ensure field feasibility, including stand-off distance for safe operation, sensor field-of-view limits, endurance, and any site-specific operational restrictions.

Based on these inputs, the framework produces two outputs that are directly relevant to maintenance-oriented inspection. The first is a set of inspection viewpoints that are explicitly tied to bridge elements and their inspection requirements, rather than being selected purely for generic coverage. The second is an executable UAV route that connects these viewpoints while respecting operational constraints. A key design requirement is repeatability: the framework is intended to generate inspection paths that remain consistent across inspection cycles unless maintenance knowledge indicates a justified change in inspection priorities. This is essential for longitudinal comparability, allowing inspection outcomes to be interpreted not as isolated observations but as part of a continuous monitoring process that informs maintenance planning over time.

3.2 Knowledge-based planning workflow

Building on the overview in Section 3.1, this subsection explains how maintenance knowledge is used to shape UAV inspection planning in a way that remains practical for field deployment and useful for maintenance decision-making. The workflow begins by translating maintenance intent into explicit inspection objectives, rather than treating the bridge as a generic surface to be covered. This emphasis reflects recent observations that UAV inspection outcomes can be inconsistent across cycles when route

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definition is largely manual or guided only by geometric convenience (Hou et al., 2022; Panigati et al., 2025).

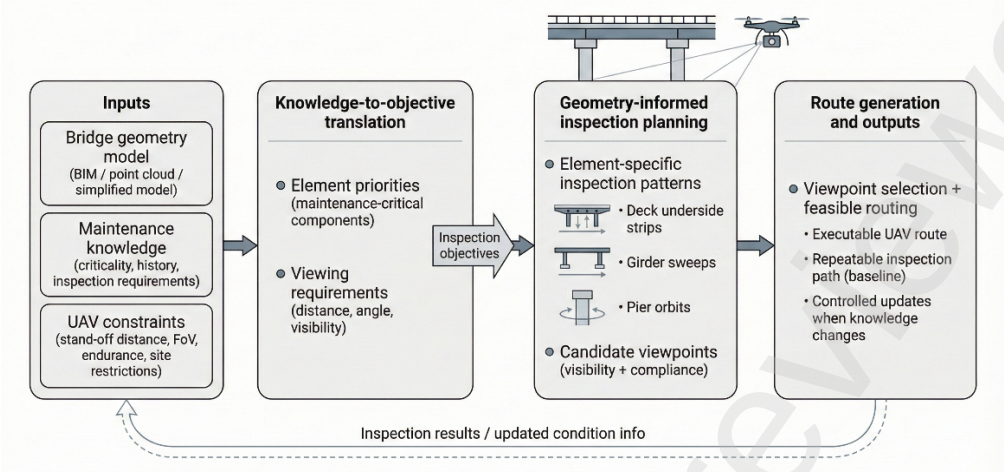


Fig. 3. Knowledge-based UAV path planning framework for bridge inspection and maintenance

The first step is to interpret maintenance knowledge into element-level priorities and inspection requirements. Each bridge element is associated with a priority level that reflects maintenance relevance, for example based on component criticality, historical condition patterns, or known vulnerability to specific deterioration mechanisms. In addition, each element is assigned basic viewing requirements that define what constitutes an acceptable inspection observation, such as a desired camera-to-surface distance, a preferred range of viewing angles, and minimum expectations for visibility and coverage. Together, these attributes determine not only which elements should be inspected first, but also how strictly the inspection should be performed for each element.

The second step uses bridge geometry to generate inspection patterns that are appropriate for different structural elements. Instead of relying on a single coverage strategy, the workflow adopts element-specific inspection logic. For instance, deck undersides are naturally suited to longitudinal or strip-like passes that maintain stable stand-off distance under shaded conditions, while girders can be inspected through guided sweeps aligned with their dominant direction. Piers and substructures may require orbital or circumferential viewpoints to ensure that occlusions are minimized and multiple faces are observed. The role of geometry in this step is therefore not limited to collision avoidance; it provides the spatial context needed to translate inspection requirements into feasible candidate viewpoints.

The third step selects candidate viewpoints and constructs an executable route under UAV operational constraints. Candidate viewpoints are evaluated based on their ability to satisfy element-specific viewing requirements while contributing to the inspection objectives defined by maintenance knowledge. The route is then formed by connecting the selected viewpoints in a feasible order that respects practical constraints such as stand-off distance, field-of-view limits, endurance, and site-specific restrictions. At this stage, efficiency remains important, but it is treated as a constraint-aware planning consideration rather than the sole objective.

IV. Results

This study adopts a methodology that links maintenance intent to executable UAV inspection routes through a knowledge-based planning workflow. The methodology is organized into sequential stages that begin with preparing bridge and maintenance inputs, proceed through objective formulation and route generation, and end with evaluation against

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practical baselines. The overall process is designed to support repeatable inspections so that outputs can be compared across cycles and used for maintenance planning.

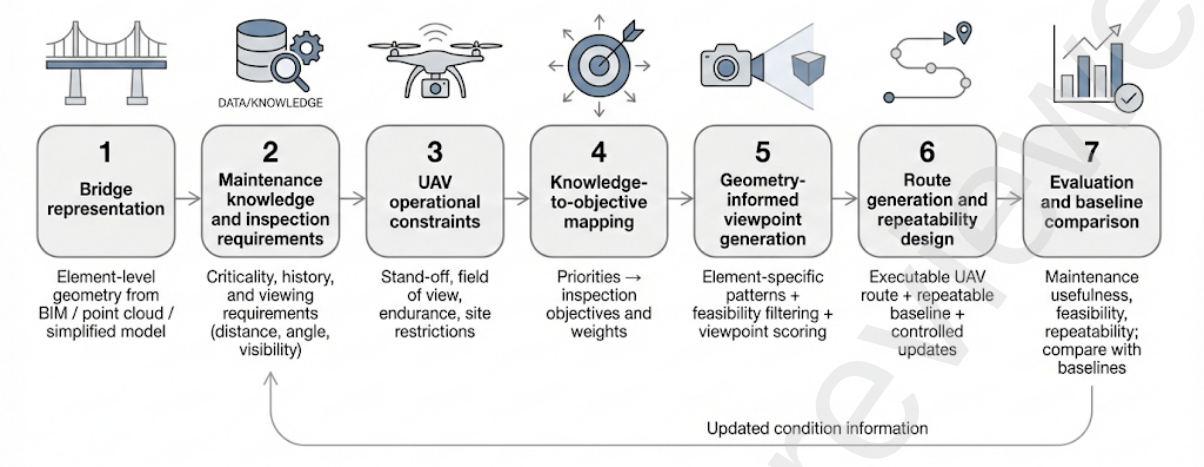


Fig. 4. Research Framework

4.1 Knowledge-to-objective mapping

Let $\mathcal{E} = \{e_1,...,e_{|\mathcal{E}|}\}$denote the set of bridge elements to be inspected (e.g., deck underside, girders, piers). Each element $e$is assigned a maintenance-informed priority weight $P_e \in [0,1]$, derived from criticality and historical condition information and updated after each inspection cycle (Zhang et al., 2023; Gao et al., 2024). A simple update rule can be written as

$$P_e^{(t+1)} = \text{clip}_{[0,1]}(\alpha P_e^{(t)} + (1-\alpha)\,\hat{P}_e^{(t)}),$$

where $\hat{P}_e^{(t)}$is a priority estimate from the latest condition evidence and $\alpha \in [0,1]$controls stability over time.

Each element is also assigned viewing requirements R_e, including a stand-off distance range $[d_e^{min}, d_e^{max}]$and a viewing angle range $[\theta_e^{min}, \theta_e^{max}]$. For a candidate viewpoint $v$with camera position x_v and element reference point x_e, define

$$d(v,e) = \| x_v - x_e \|, \theta(v,e) = \arccos\left(\frac{n_e^\top (x_e - x_v)}{\| x_e - x_v \|}\right),$$where $n_e$is the outward normal (or nominal observation direction) of element e. Distance and angle compliance can be modeled with bounded scores:

$$S_d(v,e) = \begin{cases} 1, & d_e^{min} \le d(v,e) \le d_e^{max}, \\ \exp\left(-\dfrac{|d(v,e) - \Pi_{[d_e^{min}, d_e^{max}]}(d(v,e))|}{\sigma_d}\right), & \text{otherwise}, \end{cases}$$

$$S_\theta(v,e) = \begin{cases} 1, & \theta_e^{min} \le \theta(v,e) \le \theta_e^{max}, \\ \exp\left(-\dfrac{|\theta(v,e) - \Pi_{[\theta_e^{min}, \theta_e^{max}]}(\theta(v,e))|}{\sigma_\theta}\right), & \text{otherwise}, \end{cases}$$

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where $\Pi_{[a,b]}(\cdot)$projects a value onto $[a,b]$, and $\sigma_d,\sigma_\theta > 0$are softness parameters.

Let $S_{\text{vis}}(v,e) \in \{0,1\}$indicate line-of-sight visibility (1 if unobstructed, 0 otherwise). The viewpoint quality is then

$$Q(v,e) = S_{\text{vis}}(v,e) \cdot S_d(v,e) \cdot S_\theta(v,e).$$

The maintenance-weighted inspection utility of viewpoint v becomes

$U(v) = \sum_{e \in \mathcal{E}} P_e \; Q(v,e)$, which makes maintenance intent explicit and avoids purely geometry-driven objectives (Hou et al., 2022; Panigati et al., 2025).

4.2 Viewpoint Selection and Route Generation

Candidate viewpoints are generated using bridge geometry and element-specific inspection patterns. Deck undersides are sampled through strip-like passes, girders through directional sweeps, and piers through multi-face viewpoints to reduce occlusion. Viewpoints are filtered for feasibility under UAV constraints, then associated with utility $U(v)$and travel cost $C(v_i,v_j)$. Route generation selects and orders viewpoints to maximize inspection utility under a flight budget, which is naturally framed as a prize-collecting or orienteering-type problem rather than a shortest-path task. A practical implementation uses a two-stage procedure: viewpoints are selected using a utility-to-cost criterion, then ordered using a lightweight routing heuristic to reduce travel cost while preserving inspection value. Repeatability is supported through deterministic viewpoint sampling and stable element anchors, with knowledge updates triggering local refinements rather than full route regeneration (Hou et al., 2022; Toriumi et al., 2022).

4.3 Evaluation Metrics and Baseline Comparisons

Evaluation considers maintenance usefulness alongside operational feasibility. Maintenance usefulness is measured through critical-element coverage (the proportion of high-priority elements meeting minimum viewing requirements) and viewpoint compliance (distance/angle satisfaction). Operational feasibility is assessed through path efficiency (flight distance or estimated time), while repeatability is quantified through route similarity across cycles and consistency in inspected elements and compliant viewpoints. Baselines include a manual rule-of-thumb plan, a geometry-driven coverage route without maintenance weighting, and a feasibility-oriented robust plan (e.g., wind- or GNSS-aware) that does not encode maintenance priorities. These comparisons isolate the contribution of knowledge-driven objectives, geometry-informed inspection patterns, and repeatability design (Hou et al., 2022; Panigati et al., 2025).

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V. Case Study and Experimental Setup

5.1 Study Area and Bridge Model

The case study considers a conventional **concrete box-girder highway bridge** on a toll road. The bridge is modeled at the element level using a **BIM-based geometry source**, which enables consistent identification of inspection targets and supports repeatable planning across inspection cycles. In this study, the bridge geometry is organized into maintainable components aligned with the project work breakdown structure (WBS), separating **substructure** and **superstructure** elements. This element-level representation is used to (i) define inspection elements $\mathcal{E}$, (ii) assign maintenance priorities and viewing requirements per element, and (iii) generate geometry-informed candidate viewpoints and executable UAV inspection routes.



(a) (b)

Fig. 5. (a) Field view of the concrete box-girder toll-road overpass; (b) BIM-derived bridge model

The WBS-based element grouping also provides a practical interface for maintenance teams, ensuring that the planning outputs remain traceable to maintenance packages and reporting conventions used in bridge asset management as shown as in table 2.

Table 2. Work Breakdown Structure (WBS) for concrete box-girder bridge maintenance

WBS Level 1	WBS Level 2	WBS Level 3	WBS Level 4
Concrete Bridge Maintenance Project	Substructure	Foundations	Pile Foundation
			Pile Cap
		Piers & Abutments	Pier/Abutment Wall
			Pier Head
			Pier Drainage
		Embankment Fill	Approach Embankment Fill
		Retaining Structure	Wing Wall
		Road Pavement	Flexible Pavement
	Superstructure	Girders	Approach Slab
			Main Girders (Box Girder)
			Diaphragm
			Girder Drainage
		Deck System	Deck Structure
			Deck Surfacing / Wearing Course
		Expansion Joint	Asphaltic Expansion Joint
			Elastomeric Component
		Road User Safety	Railing / Barrier System

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5.2 Maintenance Knowledge and Inspection Requirements

The maintenance knowledge used in this case study was derived from field inspection documentation and maintenance records obtained for the concrete box-girder toll-road overpass described in Section 5.1. These records were consolidated at the element level and aligned with the bridge WBS so that condition information and maintenance intent can be linked directly to executable UAV inspection planning. The resulting knowledge representation is summarized in Table 2, which reports the inspection elements $\mathcal{E}$, their priority weights, and the viewing requirements adopted for route generation.

Two complementary components are defined for each element $e \in \mathcal{E}$. The first is a maintenance-informed priority weight $P_e \in [0,1]$, which reflects how strongly the inspection should focus on element $e$from a maintenance perspective. In practical bridge maintenance settings, $P_e$is driven by factors such as component criticality, historical condition patterns, and recent inspection findings. To preserve consistency across inspection cycles while allowing priorities to respond to new evidence, the study adopts a smoothed update formulation,

$$P_e^{(t+1)} = \mathrm{clip}_{[0,1]}(\alpha P_e^{(t)} + (1-\alpha)\hat{P}_e^{(t)}),$$

where $\hat{P}_e^{(t)}$represents the evidence-informed priority estimate derived from the most recent inspection outcomes and αcontrols the stability of the prioritization over time. This design supports repeatable inspection planning while accommodating maintenance-driven changes when new condition information becomes available.

The second component is the element-specific set of viewing requirements R_e, which defines the minimum conditions for an observation to be considered inspection-valid for that element. In this study, $R_e$is specified through stand-off distance limits $[d_e^{min}, d_e^{max}]$and admissible viewing angle limits $[\theta_e^{min}, \theta_e^{max}]$, together with basic line-of-sight feasibility. These requirements are intended to reflect inspection practicality and data sufficiency. For example, under-deck and box-girder observations typically demand tighter distance control and more stable viewing angles to ensure comparable surface detail, whereas pier observations can tolerate broader viewpoints due to multi-face accessibility.

The element priorities and requirements are then mapped into the viewpoint scoring model used for planning. For a candidate viewpoint v, the element-level quality score is defined as

$$Q(v,e) = S_{\mathrm{vis}}(v,e) \cdot S_d(v,e) \cdot S_\theta(v,e),$$

where $S_d$and $S_\theta$capture compliance with the distance and angle requirements, and S_{vis}represents visibility feasibility. Viewpoint utility is aggregated across elements using the maintenance-informed priorities,

$$U(v) = \sum_{e \in \mathcal{E}} P_e \; Q(v,e),$$

so that the planning objective reflects maintenance intent rather than generic geometric coverage. Table 3 provides the element-level $P_e$and $R_e$values used in this case study and serves as the primary input to the viewpoint selection and route generation procedures described in Section 4. Data handling and traceability. The field records were mapped to BIM-defined elements to ensure that priorities and requirements remain traceable to maintainable components and to facilitate consistent route regeneration across cycles. Route outputs are exported as waypoint sequences for operational use and for repeatability assessment in subsequent inspections.

Table 3. WBS-based bridge elements and maintenance trend indicators.

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ElementID	WBS_L4	ElementType	Criticality	ConditionIndex	DefectSeverity	Slope_Severity	P_e
E12	Deck structure	deck	High	76.012	0.361	0.070	0.699
E09	Main girders (box girder)	box_girder	High	79.134	0.346	-0.016	0.589
E03	Pier/abutment wall	pier	High	74.273	0.199	0.008	0.542
E04	Pier head	pier_head	High	81.925	0.221	-0.002	0.538
E15	Elastomeric component	bearing	High	61.889	0.175	0.005	0.527
E02	Pile cap	foundation	High	78.814	0.118	0.003	0.502
E01	Pile foundation	foundation	High	88.929	0.093	-0.011	0.487
E11	Girder drainage	drainage	Medium	62.914	0.233	0.029	0.407
E06	Wing wall	retaining	Medium	86.918	0.217	0.021	0.388
E10	Diaphragm	diaphragm	Medium	72.397	0.282	-0.008	0.383
E08	Approach slab	approach_slab	Medium	83.765	0.174	0.024	0.376
E16	Railing / barrier system	railing	Medium	75.920	0.148	0.026	0.369
E14	Asphaltic expansion joint	joint	Medium	58.552	0.184	0.016	0.368
E05	Pier drainage	drainage	Medium	70.076	0.173	-0.005	0.339
E17	Lighting system	lighting	Low	76.684	0.158	0.021	0.230
E07	Flexible pavement (approach)	approach	Low	75.463	0.102	-0.002	0.176
E13	Wearing course / surfacing	surfacing	Low	87.739	0.066	0.010	0.176

In addition, the BIM model was used to visualize inspection-target elements through a priority heatmap, where the color scale transitions from green to red to indicate low to high inspection priority. The heatmap is generated (as shown as in figure 6) by mapping element-level maintenance weights (e.g., P_e) onto the corresponding BIM components, so that the bridge's inspection intent becomes explicit at the geometric level rather than remaining as a purely tabular attribute. This visualization serves as a practical sanity check that the prioritization logic is consistently linked to the correct structural elements, particularly for components that are visually similar or spatially adjacent in the superstructure and substructure. It also provides a shared visual language for stakeholders, enabling engineers, UAV operators, and maintenance planners to interpret the intended inspection emphasis in the same way, verify that critical components receive adequate attention, and communicate route planning outputs more consistently across inspection cycles.

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Fig. 6. BIM visualization of inspection priorities (green = low, red = high).

5.3 UAV Platform and Operational Constraints

To ensure that the generated inspection routes remain executable in real bridge environments, this study defines UAV capabilities and operational constraints that bound both viewpoint feasibility and route construction. The UAV is treated as a multirotor inspection platform equipped with a stabilized RGB camera, reflecting typical bridge inspection practice where hovering and controlled stand-off distances are required. Rather than binding the method to a single manufacturer model, constraints are expressed in a platform-agnostic form, so that the framework can be reproduced for different UAVs provided comparable sensing and endurance envelopes are available.

Operational constraints are imposed in three layers: sensing validity, safety feasibility, and mission budget. At the sensing level, each candidate viewpoint $v$must satisfy element-specific requirements R_e defined in Table 2, particularly the admissible stand-off distance $[d_e^{min},d_e^{max}]$and viewing angle range $[\theta_e^{min},\theta_e^{max}]$. At the safety level, the viewpoint must also respect a minimum clearance from the bridge geometry, expressed as

$$d(v,\Gamma) \geq d_{\text{safe}},$$

where Γdenotes the bridge surface model and $d_{\text{safe}}$is the prescribed operational clearance. Viewpoints outside allowed flight regions (e.g., exclusion buffers above traffic lanes or restricted areas) are removed using a hard feasibility filter $g_k(v) \leq 0$for all relevant constraints k.

At the route level, feasibility is governed by a mission budget that approximates endurance limits. For a route $\pi = \{v_1,...,v_m\}$, the travel cost is defined as

$$\text{Cost}(\pi) = \sum_{i=0}^{m-1} C(v_i,v_{i+1}),v_0 = \text{HOME},$$

with a corresponding budget constraint $\text{Cost}(\pi) \leq B$. In the base formulation, the transition cost can be expressed as Euclidean travel distance,

$$C(v_i,v_j) = \| x_{v_i} - x_{v_j} \|,$$

or converted to time using nominal cruise speed $\grave{s}$, $C_t(v_i,v_j) = C(v_i,v_j)/\grave{s}$. These constraints enforce that the selected viewpoints yield a route that is not only inspection-valid but also realistically flyable within battery and operational limits.

To reflect common field challenges around bridges, a robust feasibility variant is also considered as a baseline, where transitions are penalized in operationally unfavorable regions such as GNSS-

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shadowed under-deck spaces or wind-exposed corridors near deck edges. In this case, the transition cost is modified as

$$C_{\text{rob}}(v_i,v_j) = C(v_i,v_j)(1 + \lambda\,\psi(v_i,v_j)),$$

where $\psi(\cdot)$is a bounded penalty function and λcontrols its influence. This design captures the practical reality that routes may remain feasible in pure geometric terms but still be undesirable due to navigation robustness or environmental exposure.

The impact of these operational assumptions is illustrated by comparing the planned routes across the proposed method and baseline strategies. Figure 7 visualizes four representative routes: a manual rule-of-thumb route, a coverage-only baseline, a robust feasibility baseline, and the proposed knowledge-based route. While the baselines tend to reflect either ad hoc operational choices (manual) or generic geometric preferences (coverage-only), the proposed route is shaped by maintenance-weighted inspection value under the same mission budget, resulting in a pattern that remains comparable across cycles while explicitly emphasizing maintenance-critical regions. In the next subsection, these route differences are quantified using coverage, compliance, efficiency, and repeatability metrics.

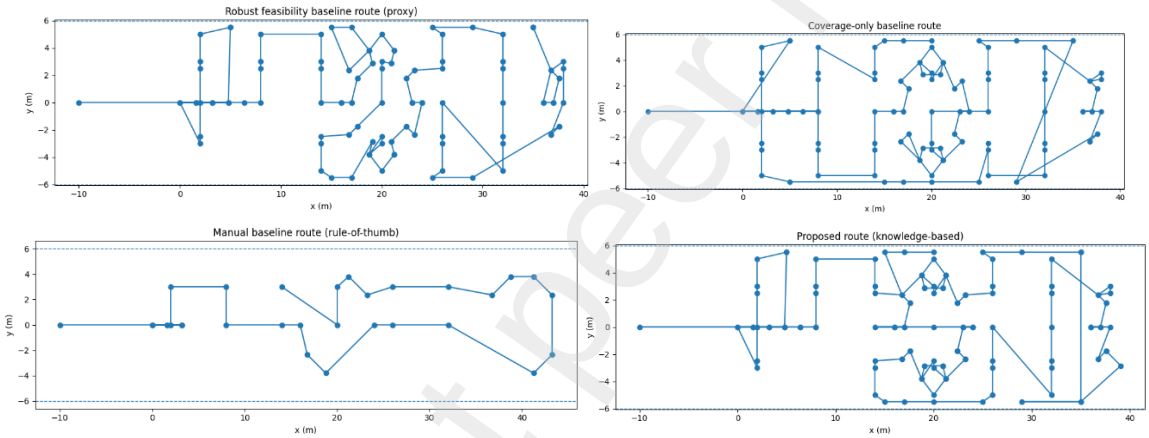


Fig. 7. comparison of UAV inspection routes

To make the comparison between the proposed approach and the baselines explicit, each route generator can be described using a common budgeted selection form, where a route $\pi \subseteq \mathcal{V}$is constructed from candidate viewpoints $\mathcal{V}$under a mission budget B. In the coverage-only baseline, the objective is to maximize generic inspection utility without maintenance weighting. Let $\grave{Q}(v)$denote the mean element quality at viewpoint v, computed as

$$\grave{Q}(v) = \frac{1}{|\mathcal{E}|}\sum_{e\in\mathcal{E}} Q(v,e).$$

The coverage-only baseline can then be expressed as

$$\max_{\pi\subseteq\mathcal{V}} \sum_{v\in\pi} \grave{Q}\,(v)\text{s.t.}\text{Cost}(\pi) \le B.$$

This formulation encourages routes that perform well on average across elements but does not encode maintenance intent, so high-value maintenance targets are not necessarily emphasized when trade-offs arise.

The robust feasibility baseline follows the same generic coverage objective but modifies the budget constraint through a robustness-aware travel cost to proxy navigation and environmental challenges

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around bridges. Using the penalized transition cost $C_{\text{rob}}(\cdot)$defined previously, the robust baseline is written as

$$\max_{\pi \subseteq \mathcal{V}} \sum_{v \in \pi} \bar{Q}(v) \text{s.t.} \text{Cost}_{\text{rob}}(\pi) = \sum_{i=0}^{m-1} C_{\text{rob}}(v_i, v_{i+1}) \leq B.$$

In this case, the route is biased away from proxy "risk zones" by inflating travel cost, but the inspection objective remains maintenance-agnostic.

The manual rule-of-thumb baseline represents common practice where the route is defined heuristically to cover major components using operator judgment rather than an explicit objective function. It can be interpreted as selecting a predefined subset π_{man}based on simple rules (e.g., "visit deck underside strips, at least one pier orbit, and one pass near joints") and then ordering the selected points using a short routing heuristic. While this baseline is operationally plausible, it typically lacks a consistent mapping from maintenance knowledge to viewpoint choice, which can reduce comparability across inspection cycles.

In contrast, the proposed knowledge-based method replaces the generic coverage score with maintenance-weighted inspection utility $U(v)$, where

$$U(v) = \sum_{e \in \mathcal{E}} P_e \, Q(v,e).$$

The planning objective is then

$$\max_{\pi \subseteq \mathcal{V}} \sum_{v \in \pi} U(v) \text{ s.t.} \text{Cost}(\pi) \leq B,$$

which ensures that viewpoint selection and route construction are driven by element-level maintenance priorities under the same feasibility constraints. This explicit formulation clarifies that the key difference is not route optimization alone, but the integration of maintenance intent into the objective function, which is expected to improve maintenance-critical coverage and cycle-to-cycle repeatability.

Taken together, these formulations provide a unified basis for comparison: all generators operate on the same candidate viewpoint set $\mathcal{V}$and are constrained by an equivalent mission budget B, while differing primarily in how inspection value is defined and how feasibility is encoded. This makes it possible to attribute performance differences to the presence (or absence) of maintenance intent in the planning objective, rather than to unequal operational assumptions. In particular, the coverage-only and robust feasibility baselines reflect two common priorities in practice—maximizing generic inspection usefulness and improving operational robustness—yet both remain maintenance-agnostic by construction. The manual baseline, while field-realistic, illustrates how heuristic planning can introduce variability that limits cross-cycle comparability. Against this backdrop, the proposed method is evaluated as a mechanism for translating element-level maintenance priorities into route-level decisions under the same constraints, with the expectation that improvements will be most visible in maintenance-critical coverage and repeatability metrics reported in the subsequent evaluation.

VI. Results & Discussion

This section evaluates whether translating maintenance intent into route-level decisions leads to inspection plans that are not only feasible, but also more useful for maintenance-oriented monitoring. The comparison uses the same candidate viewpoint pool $\mathcal{V}$, the same mission budget B, and the same feasibility constraints across methods. The only deliberate difference is the planning objective and cost model (Table X), which allows performance differences to be interpreted as the effect of maintenance

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knowledge integration rather than unequal operational assumptions. The resulting route patterns are visualized in Figure 7, which contrasts the proposed knowledge-based plan with three baselines: manual rule-of-thumb, coverage-only, and a robust feasibility proxy.

Figure 7 shows that the manual baseline produces a plausible route but tends to reflect operator heuristics (e.g., "hit the main components once"), which can make emphasis and viewpoint geometry drift across cycles as site conditions and operator judgment change. This limitation aligns with the broader observation in drone bridge inspection reviews that operational execution variability (distance, angle, lighting, and path repeatability) strongly influences whether UAV imagery can support reliable condition tracking over time (Toriumi et al., 2022; Panigati et al., 2025). The coverage-only baseline, by construction, favors viewpoints that perform reasonably well on average, but it does not consistently "spend" the flight budget where maintenance decision-making is most sensitive. Similar tendencies can be observed in geometry-driven or efficiency-driven coverage planning studies, where viewpoint redundancy reduction and maneuver efficiency are the primary targets, while maintenance intent remains implicit (Li et al., 2025). The robust feasibility baseline improves operational realism by discouraging transitions through proxy risk zones, reflecting concerns that under-deck or edge-exposed regions can be operationally challenging. This is consistent with recent bridge inspection planning work that explicitly considers wind effects or GNSS shadowing to improve feasibility (Xu & Dai, 2024; Kim et al., 2025). However, feasibility-oriented robustness alone does not guarantee that the best viewpoints for maintenance-critical components are selected when trade-offs arise under limited endurance.

Against these baselines, the proposed method visibly concentrates viewpoint allocation around maintenance-priority elements while preserving a flyable route within the same budget (Figure 7). This behavior is expected from the objective $\sum_{v\in\pi} U(v)$with $U(v) = \sum_{e\in\mathcal{E}} P_e Q(v,e)$, which explicitly rewards viewpoints that satisfy viewing requirements for high-priority elements. In practical terms, the route is less about "maximum surface coverage" and more about achieving repeatable, inspection-valid observations where maintenance decisions are most affected. This emphasis is aligned with the growing direction of integrating inspection activities into broader asset management and digital twin workflows (Bono et al., 2022), but with a clearer operational bridge between maintenance knowledge and executable route generation.

Table 4. Quantitative Comparison of Route-Planning Methods

Method	CEC	VRC	Cost (m)	Cost (min)	Robust cost (m)	#Views	# High-priority elements
Proposed (knowledge-based)	1.00	0.157	288.4	1.60	355.7	121	5
Manual (rule-of-thumb)	1.00	0.164	116.5	0.65	147.7	30	5
Coverage-only	1.00	0.151	303.2	1.68	368.9	126	5
Robust feasibility (proxy)	1.00	0.166	260.3	1.45	323.1	114	5

To make the comparison auditable, the study reports four complementary metric groups in Table 4 (recommended): (i) maintenance relevance through critical element coverage (CEC), (ii) data quality feasibility through viewpoint requirement compliance (VRC), (iii) operational efficiency through total distance/time, and (iv) repeatability using a spatial similarity score across cycles (e.g., average nearest-neighbor distance between waypoint sets and overlap in inspected elements). The intention is to separate

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"being able to fly" from "collecting comparable and maintenance-useful observations," which is a recurring gap noted in UAV inspection practice reviews (Panigati et al., 2025; Toriumi et al., 2022). In many geometry-driven planning approaches, better efficiency can be achieved by reducing redundant viewpoints and sharp maneuvers, but the metrics most directly tied to maintenance interpretation—such as consistent compliance with viewpoint requirements for the same elements across cycles—are not always foregrounded (Li et al., 2025). This is precisely where a maintenance-weighted objective is expected to show value: not necessarily by producing the shortest route, but by improving maintenance-critical coverage and the repeatability of observation geometry under the same budget.

To help readers quickly interpret trade-offs, a compact visualization such as Figure 7 (recommended) can summarize Table 4 using a bar chart comparing methods on CEC, VRC, and cost. This style of reporting is consistent with recent tool-focused and workflow-oriented contributions that stress operational comparability and quality control across UAS inspection deployments (Bartczak et al., 2025). If you include the robust baseline, it is useful to report both standard cost and robustness-penalized cost (as implemented in your Colab), because it clarifies whether "robustness" is gained by simply avoiding risky regions at the expense of maintenance relevance.

Recent studies increasingly move beyond "UAVs can inspect bridges" toward "how UAV inspection can be operationalized as a repeatable program." Reviews emphasize that adoption barriers are often practical—standardization, comparability, and integration with decision processes—rather than purely sensing capability (Bolourian, 2020; Bono, 2022). Methodologically, several strands address parts of this puzzle. Coverage path planning and viewpoint optimization reduce redundancy and improve flight smoothness (Li et al., 2022), wind-aware planning improves realism in exposed environments (Xu & Dai, 2024), and GNSS-shadow-aware substructure planning improves feasibility under decks (Kim et al., 2023). Digital-twin-linked approaches define reference trajectories and connect UAV inspection to bridge management workflows, which strengthens integration at the system level (Bono et al., 2022). Practical flight planning toolkits also show that "optimized routing" can be embedded in repeatable workflows when geometry is available (Fatemidokht et al., 2021). At the same time, adaptive inspection strategies increasingly exploit prior information to trigger detailed inspections in areas likely to contain defects (Chen et al., 2023), but these approaches still often treat maintenance knowledge as an auxiliary cue rather than a formal objective that controls viewpoint validity and cross-cycle repeatability.

Within this landscape, the distinguishing contribution demonstrated here is the explicit mapping of maintenance intent into inspection objectives and then into executable routing decisions under a shared budget. The proposed method is not positioned as replacing robustness- or coverage-focused advances; instead, it complements them by making "maintenance relevance" the primary planning signal while keeping feasibility filters compatible with wind/GNSS or other operational constraints. This integration is important because bridge maintenance decisions depend on consistent, comparable observations of the same critical elements over time, not simply on the availability of UAV imagery. The heatmap visualization in the BIM environment (Figure 6) further supports this positioning by showing that the element priorities driving the route are traceably mapped onto geometry, enabling transparent communication between maintenance engineers and UAV operators.

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VII. Conclusion

This paper presented a knowledge-based approach to UAV path planning for bridge inspection and maintenance, motivated by the practical need to make UAV inspections not only feasible, but also repeatable and directly useful for maintenance decision-making. The key idea is to translate maintenance intent into explicit inspection objectives by assigning element-level priorities and viewing requirements, and then to generate executable routes by selecting and ordering candidate viewpoints under a mission budget. By embedding maintenance weighting into the planning objective, the framework shifts route design from generic surface coverage toward inspection plans that target maintenance-critical components with inspection-valid viewpoints.

The case study on a conventional concrete box-girder toll-road bridge demonstrated that the proposed planner can operate within standard UAV operational constraints while producing routes that are clearly distinguishable from manual and geometry-driven baselines. Under the same budget assumptions, all compared methods achieved full coverage of high-priority elements (CEC = 1.0), indicating that the bridge-scale inspection scope was achievable across strategies. Differences emerged more clearly in route structure and compliance-oriented indicators, where the proposed method provides a transparent mechanism to maintain consistent inspection intent across cycles rather than relying on operator heuristics or average-coverage optimization. The integration of BIM-based visualization further strengthened traceability by confirming that element priorities driving the route are correctly mapped onto the bridge geometry, enabling clearer communication between maintenance teams and UAV operators.

From a research perspective, the contribution of this work lies in making the maintenance-to-execution link explicit: maintenance knowledge is operationalized as a route planning objective, rather than remaining a post-processing or decision-support artifact. This positioning complements recent advances in geometry-driven coverage optimization and robustness-aware navigation by providing a planning layer that can incorporate those feasibility improvements while preserving maintenance relevance. In practical deployments, this approach supports continuous inspection by enabling comparable viewpoints over time and allowing routes to adapt selectively when maintenance priorities change, rather than requiring complete re-planning from scratch.

Several limitations provide clear directions for future work. First, the current feasibility modeling uses simplified proxies for environmental and navigation effects; integration with higher-fidelity wind fields, GNSS-denied localization models, and obstacle-aware planning would strengthen operational realism. Second, viewpoint validity is defined through interpretable distance and angle requirements, but future studies should link these criteria more directly to measurable data quality outcomes such as crack detectability limits, photogrammetric reconstruction accuracy, or uncertainty in defect quantification. Third, although element priorities were derived from maintenance records and mapped to BIM, broader validation across multiple bridge types and inspection regimes is needed to establish generalizability and to refine how priorities and requirements should be standardized across agencies.

Overall, the proposed knowledge-based planning framework provides a practical step toward maintenance-oriented UAV inspection programs, where flight routes are not only flyable, but also repeatable, traceable to maintenance intent, and suitable for long-term condition tracking and maintenance planning.

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VIII. Declaration

Author Contributions

The following statements should be used:

Conceptualization, **N.B.L.** and **Y.L.**; methodology, **N.B.L.**; software, **N.B.L.**; validation, **N.B.L., Y.L., and B.T.**; formal analysis, **N.B.L.**; investigation, **N.B.L.**; resources, **Y.L. and B.T.**; data curation, **N.B.L.**; writing—original draft preparation, **N.B.L.**; writing—review and editing, **Y.L., B.T., K.S., and N.S.**; visualization, **N.B.L.**; supervision, **Y.L., B.T., and K.S.**; project administration, **Y.L.**; funding acquisition, **Y.L. and B.T.**

Data Availability Statement

The data presented in this study are available from the corresponding author upon reasonable request. The data are not publicly available due to confidentiality, privacy, and ethical restrictions associated with the project.

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Conflicts of Interest

The authors declare no conflict of interest.

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References

Aela, P. (2024). UAV-based studies in railway infrastructure monitoring. *Automation in Construction*, *167*(Query date: 2026-01-27 11:49:48). https://doi.org/10.1016/j.autcon.2024.105714

Agredo Chávez, A., Gonzalez-Libreros, J., Wang, C., & Sas, G. (2024). Bridge condition index: A review of methodologies used in Bridge Management Systems. In Jensen J.S., Frangopol D.M., & Schmidt J.W. (Eds.), *Bridge Maint., Saf., Manag., Digit. Sustain. - Proc. Int. Conf. Bridge Maint., Saf. Manag.* (pp. 1130–1137). CRC Press/Balkema. Scopus. https://doi.org/10.1201/9781003483755-131

Aliyari, M. (2022). Hazards identification and risk assessment for UAV–assisted bridge inspections. *Structure and Infrastructure Engineering*, *18*(3), 412–428. https://doi.org/10.1080/15732479.2020.1858878

Ameli, Z. (2022). Impact of UAV Hardware Options on Bridge Inspection Mission Capabilities. *Drones*, *6*(3). https://doi.org/10.3390/drones6030064

Ayele, Y. Z. (2020). Automatic crack segmentation for uav-assisted bridge inspection. *Energies*, *13*(23). https://doi.org/10.3390/en13236250

Biscarini, C. (2020). UAV photogrammetry, infrared thermography and GPR for enhancing structural and material degradation evaluation of the Roman masonry bridge of Ponte Lucano in Italy. *NDT and E International*, *115*(Query date: 2026-01-27 11:49:48). https://doi.org/10.1016/j.ndteint.2020.102287

Bolourian, N. (2020). LiDAR-equipped UAV path planning considering potential locations of defects for bridge inspection. *Automation in Construction*, *117*(Query date: 2026-01-27 11:49:48). https://doi.org/10.1016/j.autcon.2020.103250

Bono, A. (2022). Path Planning and Control of a UAV Fleet in Bridge Management Systems. *Remote Sensing*, *14*(8). https://doi.org/10.3390/rs14081858

Chen, J.-H., Su, M.-C., Lin, S.-K., Lin, W.-J., & Gheisari, M. (2023). Smart bridge maintenance using cluster merging algorithm based on self-organizing map optimization. *Automation in Construction*, *152*. Scopus. https://doi.org/10.1016/j.autcon.2023.104913

Falorca, J. F. (2021). New trends in visual inspection of buildings and structures: Study for the use of drones. *Open Engineering*, *11*(1), 734–743. https://doi.org/10.1515/eng-2021-0071

Fang, Z. (2024). Strategies for Optimized UAV Surveillance in Various Tasks and Scenarios: A Review. *Drones*, *8*(5). https://doi.org/10.3390/drones8050193

Fatemidokht, H., Rafsanjani, M., & ... (2021). Efficient and secure routing protocol based on artificial intelligence algorithms with UAV-assisted for vehicular ad hoc networks in intelligent transportation systems. *... on Intelligent ...*, (Query date: 2024-04-12 04:23:52). https://ieeexplore.ieee.org/abstract/document/9312485/

Gao, Y., Xiong, G., Li, H., & Richards, J. (2024). Exploring bridge maintenance knowledge graph by leveraging GrapshSAGE and text encoding. *Automation in Construction*, *166*. Scopus. https://doi.org/10.1016/j.autcon.2024.105634

Gucunski, N., Kee, S., La, H., Basily, B., & ... (2015). Delamination and concrete quality assessment of concrete bridge decks using a fully autonomous RABIT platform. *... and Maintenance*, (Query date: 2024-09-25 19:03:28). https://www.researchgate.net/profile/Hung-La-3/publication/276088804_Delamination_and_concrete_quality_assessment_of_concrete_bridge_decks_using_a_fully_autonomous_RABIT_platform/links/55fcfc3908aeba1d9f4b7d54/Delamination-and-concrete-quality-assessment-of-concrete-bridge-decks-using-a-fully-autonomous-RABIT-platform.pdf?_sg%5B0%5D=started_experiment_milestone&origin=journalDetail

Hildmann, H., & Kovacs, E. (2019). Using unmanned aerial vehicles (UAVs) as mobile sensing platforms (MSPs) for disaster response, civil security and public safety. *Drones*, (Query date: 2024-04-12 04:23:52). https://www.mdpi.com/2504-446X/3/3/59

Hoang, Q. (2021). *Using Conceptual Graphs to build failure knowledge model for forensic in civil engineering*. hal.univ-smb.fr. https://hal.univ-smb.fr/U-BORDEAUX/tel-03282455

Hu, D., Yee, T., & Goff, D. (2024). Automated crack detection and mapping of bridge decks using deep learning and drones. *Journal of Civil Structural Health Monitoring*, *14*(3), 729–743. https://doi.org/10.1007/s13349-023-00750-0

Kao, S. P. (2023). Combining the YOLOv4 Deep Learning Model with UAV Imagery Processing Technology in the Extraction and Quantization of Cracks in Bridges. *Sensors*, *23*(5). https://doi.org/10.3390/s23052572

Kim, S., Kwon, D., Jang, A., Ju, Y., Lee, J., & Hong, S. (2023). A review of UAV integration in forensic civil engineering: From sensor technologies to geotechnical, structural and water infrastructure applications. *Measurement*, (Query date: 2024-07-11 13:06:53). https://www.sciencedirect.com/science/article/pii/S0263224123014501

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=6239751

Li, C., Guo, Z., Su, D., Xiao, B., & Tam, V. (2022). The Application of Advanced Information Technologies in Civil Infrastructure Construction and Maintenance. *Sustainability*, (Query date: 2024-04-09 06:08:55). https://www.mdpi.com/2071-1050/14/13/7761

Luciw, M., Olivera, S., Gorshechnikov, A., Wurbs, J., & ... (2024). Systems and methods to enable continual, memory-bounded learning in artificial intelligence and deep learning continuously operating applications across networked …. *US Patent …*, (Query date: 2024-04-12 04:23:52). https://patents.google.com/patent/US11928602B2/en

Mishra, M., Lourenço, P., & Ramana, G. (2022). Structural health monitoring of civil engineering structures by using the internet of things: A review. *Journal of Building Engineering*, (Query date: 2024-04-09 06:08:55). https://www.sciencedirect.com/science/article/pii/S235271022101812X

Omar, T., & Nehdi, M. L. (n.d.). *Thermal Detection of Subsurface Delaminations in Reinforced Concrete Bridge Decks Using Unmanned Aerial Vehicle*. Retrieved September 10, 2024, from https://www.concrete.org/publications/internationalconcreteabstractsportal.aspx?m=details&id=51715590

Panigati, T., Zini, M., Striccoli, D., Giordano, P. F., Tonelli, D., Limongelli, M. P., & Zonta, D. (2025). Drone-based bridge inspections: Current practices and future directions. *Automation in Construction*, *173*, 106101. https://doi.org/10.1016/j.autcon.2025.106101

Ri, S. (2024). Drone-based displacement measurement of infrastructures utilizing phase information. *Nature Communications*, *15*(1). https://doi.org/10.1038/s41467-023-44649-2

Shults, R., Ormambekova, A., Medvedskij, Y., & Annenkov, A. (2023). GNSS-Assisted Low-Cost Vision-Based Observation System for Deformation Monitoring. *Applied Sciences*, *13*(5), 2813. Publicly Available Content Database (2785182787). https://doi.org/10.3390/app13052813

Wang, F. (2024). Automated UAV path-planning for high-quality photogrammetric 3D bridge reconstruction. *Structure and Infrastructure Engineering*, *20*(10), 1595–1614. https://doi.org/10.1080/15732479.2022.2152840

Wang, Y., Xiong, W., Zhu, Y., & Cai, C. S. (2026). Knowledge graph-driven bridge maintenance decision-making via integrating large language models and chain-of-thought reasoning. *Automation in Construction*, *181*, 106632. https://doi.org/10.1016/j.autcon.2025.106632

Xu, S. (2021). Analysis of Temperature and Humidity Distribution and Corrosion Effect on Suspender of Arch Bridge. In Yang M., Lanzinha J.C.G., & Bao X. (Eds.), *Adv. Transdiscipl. Eng.* (Vol. 19, pp. 262–268). IOS Press BV. Scopus. https://doi.org/10.3233/ATDE210176

Ye, S. (2020). Technology for condition and performance evaluation of highway bridges. *Journal of Civil Structural Health Monitoring*, *10*(4), 573–594. https://doi.org/10.1007/s13349-020-00403-6

Ye, X. W. (2022). Post-earthquake damage recognition and condition assessment of bridges using UAV integrated with deep learning approach. *Structural Control and Health Monitoring*, *29*(12). https://doi.org/10.1002/stc.3128

Yeum, C. M., & Dyke, S. J. (2015). Vision-Based Automated Crack Detection for Bridge Inspection. *Computer-Aided Civil and Infrastructure Engineering*, *30*(10), 759–770. https://doi.org/10.1111/mice.12141

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The following statements should be used:

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1.2. Data Availability Statement

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1.5. Conflicts of Interest

The authors declare no conflict of interest.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=6239751

References

Aela, P. (2024). UAV-based studies in railway infrastructure monitoring. *Automation in Construction*, *167*(Query date: 2026-01-27 11:49:48). https://doi.org/10.1016/j.autcon.2024.105714

Agredo Chávez, A., Gonzalez-Libreros, J., Wang, C., & Sas, G. (2024). Bridge condition index: A review of methodologies used in Bridge Management Systems. In Jensen J.S., Frangopol D.M., & Schmidt J.W. (Eds.), *Bridge Maint., Saf., Manag., Digit. Sustain. - Proc. Int. Conf. Bridge Maint., Saf. Manag.* (pp. 1130–1137). CRC Press/Balkema. Scopus. https://doi.org/10.1201/9781003483755-131

Aliyari, M. (2022). Hazards identification and risk assessment for UAV–assisted bridge inspections. *Structure and Infrastructure Engineering*, *18*(3), 412–428. https://doi.org/10.1080/15732479.2020.1858878

Ameli, Z. (2022). Impact of UAV Hardware Options on Bridge Inspection Mission Capabilities. *Drones*, *6*(3). https://doi.org/10.3390/drones6030064

Ayele, Y. Z. (2020). Automatic crack segmentation for uav-assisted bridge inspection. *Energies*, *13*(23). https://doi.org/10.3390/en13236250

Biscarini, C. (2020). UAV photogrammetry, infrared thermography and GPR for enhancing structural and material degradation evaluation of the Roman masonry bridge of Ponte Lucano in Italy. *NDT and E International*, *115*(Query date: 2026-01-27 11:49:48). https://doi.org/10.1016/j.ndteint.2020.102287

Bolourian, N. (2020). LiDAR-equipped UAV path planning considering potential locations of defects for bridge inspection. *Automation in Construction*, *117*(Query date: 2026-01-27 11:49:48). https://doi.org/10.1016/j.autcon.2020.103250

Bono, A. (2022). Path Planning and Control of a UAV Fleet in Bridge Management Systems. *Remote Sensing*, *14*(8). https://doi.org/10.3390/rs14081858

Chen, J.-H., Su, M.-C., Lin, S.-K., Lin, W.-J., & Gheisari, M. (2023). Smart bridge maintenance using cluster merging algorithm based on self-organizing map optimization. *Automation in Construction*, *152*. Scopus. https://doi.org/10.1016/j.autcon.2023.104913

Falorca, J. F. (2021). New trends in visual inspection of buildings and structures: Study for the use of drones. *Open Engineering*, *11*(1), 734–743. https://doi.org/10.1515/eng-2021-0071

Fang, Z. (2024). Strategies for Optimized UAV Surveillance in Various Tasks and Scenarios: A Review. *Drones*, *8*(5). https://doi.org/10.3390/drones8050193

Fatemidokht, H., Rafsanjani, M., & ... (2021). Efficient and secure routing protocol based on artificial intelligence algorithms with UAV-assisted for vehicular ad hoc networks in intelligent transportation systems. *... on Intelligent ...*, (Query date: 2024-04-12 04:23:52). https://ieeexplore.ieee.org/abstract/document/9312485/

Gao, Y., Xiong, G., Li, H., & Richards, J. (2024). Exploring bridge maintenance knowledge graph by leveraging GrapshSAGE and text encoding. *Automation in Construction*, *166*. Scopus. https://doi.org/10.1016/j.autcon.2024.105634

Gucunski, N., Kee, S., La, H., Basily, B., & ... (2015). Delamination and concrete quality assessment of concrete bridge decks using a fully autonomous RABIT platform. *... and Maintenance*, (Query date: 2024-09-25 19:03:28). https://www.researchgate.net/profile/Hung-La-3/publication/276088804_Delamination_and_concrete_quality_assessment_of_concrete_bridge_decks_using_a_fully_autonomous_RABIT_platform/links/55fcfc3908aeba1d9f4b7d54/Delamination-and-concrete-quality-assessment-of-concrete-bridge-decks-using-a-fully-autonomous-RABIT-platform.pdf?_sg%5B0%5D=started_experiment_milestone&origin=journalDetail

Hildmann, H., & Kovacs, E. (2019). Using unmanned aerial vehicles (UAVs) as mobile sensing platforms (MSPs) for disaster response, civil security and public safety. *Drones*, (Query date: 2024-04-12 04:23:52). https://www.mdpi.com/2504-446X/3/3/59

Hoang, Q. (2021). *Using Conceptual Graphs to build failure knowledge model for forensic in civil engineering*. hal.univ-smb.fr. https://hal.univ-smb.fr/U-BORDEAUX/tel-03282455

Hu, D., Yee, T., & Goff, D. (2024). Automated crack detection and mapping of bridge decks using deep learning and drones. *Journal of Civil Structural Health Monitoring*, *14*(3), 729–743. https://doi.org/10.1007/s13349-023-00750-0

Kao, S. P. (2023). Combining the YOLOv4 Deep Learning Model with UAV Imagery Processing Technology in the Extraction and Quantization of Cracks in Bridges. *Sensors*, *23*(5). https://doi.org/10.3390/s23052572

Kim, S., Kwon, D., Jang, A., Ju, Y., Lee, J., & Hong, S. (2023). A review of UAV integration in forensic civil engineering: From sensor technologies to geotechnical, structural and water infrastructure applications. *Measurement*, (Query date: 2024-07-11 13:06:53). https://www.sciencedirect.com/science/article/pii/S0263224123014501

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=6239751

Li, C., Guo, Z., Su, D., Xiao, B., & Tam, V. (2022). The Application of Advanced Information Technologies in Civil Infrastructure Construction and Maintenance. *Sustainability*, (Query date: 2024-04-09 06:08:55). https://www.mdpi.com/2071-1050/14/13/7761

Luciw, M., Olivera, S., Gorshechnikov, A., Wurbs, J., & ... (2024). Systems and methods to enable continual, memory-bounded learning in artificial intelligence and deep learning continuously operating applications across networked …. *US Patent …*, (Query date: 2024-04-12 04:23:52). https://patents.google.com/patent/US11928602B2/en

Mishra, M., Lourenço, P., & Ramana, G. (2022). Structural health monitoring of civil engineering structures by using the internet of things: A review. *Journal of Building Engineering*, (Query date: 2024-04-09 06:08:55). https://www.sciencedirect.com/science/article/pii/S235271022101812X

Omar, T., & Nehdi, M. L. (n.d.). *Thermal Detection of Subsurface Delaminations in Reinforced Concrete Bridge Decks Using Unmanned Aerial Vehicle*. Retrieved September 10, 2024, from https://www.concrete.org/publications/internationalconcreteabstractsportal.aspx?m=details&id=51715590

Panigati, T., Zini, M., Striccoli, D., Giordano, P. F., Tonelli, D., Limongelli, M. P., & Zonta, D. (2025). Drone-based bridge inspections: Current practices and future directions. *Automation in Construction*, *173*, 106101. https://doi.org/10.1016/j.autcon.2025.106101

Ri, S. (2024). Drone-based displacement measurement of infrastructures utilizing phase information. *Nature Communications*, *15*(1). https://doi.org/10.1038/s41467-023-44649-2

Shults, R., Ormambekova, A., Medvedskij, Y., & Annenkov, A. (2023). GNSS-Assisted Low-Cost Vision-Based Observation System for Deformation Monitoring. *Applied Sciences*, *13*(5), 2813. Publicly Available Content Database (2785182787). https://doi.org/10.3390/app13052813

Wang, F. (2024). Automated UAV path-planning for high-quality photogrammetric 3D bridge reconstruction. *Structure and Infrastructure Engineering*, *20*(10), 1595–1614. https://doi.org/10.1080/15732479.2022.2152840

Wang, Y., Xiong, W., Zhu, Y., & Cai, C. S. (2026). Knowledge graph-driven bridge maintenance decision-making via integrating large language models and chain-of-thought reasoning. *Automation in Construction*, *181*, 106632. https://doi.org/10.1016/j.autcon.2025.106632

Xu, S. (2021). Analysis of Temperature and Humidity Distribution and Corrosion Effect on Suspender of Arch Bridge. In Yang M., Lanzinha J.C.G., & Bao X. (Eds.), *Adv. Transdiscipl. Eng.* (Vol. 19, pp. 262–268). IOS Press BV. Scopus. https://doi.org/10.3233/ATDE210176

Ye, S. (2020). Technology for condition and performance evaluation of highway bridges. *Journal of Civil Structural Health Monitoring*, *10*(4), 573–594. https://doi.org/10.1007/s13349-020-00403-6

Ye, X. W. (2022). Post-earthquake damage recognition and condition assessment of bridges using UAV integrated with deep learning approach. *Structural Control and Health Monitoring*, *29*(12). https://doi.org/10.1002/stc.3128

Yeum, C. M., & Dyke, S. J. (2015). Vision-Based Automated Crack Detection for Bridge Inspection. *Computer-Aided Civil and Infrastructure Engineering*, *30*(10), 759–770. https://doi.org/10.1111/mice.12141

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This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=6239751